

Tool support for software maintenance (Scenario #3)

We kindly request your input to explore a scenario that illustrates how tools can be used to assist code refactoring. This questionnaire is organized in three sections. The first section is related to the respondent's consent, and briefly describes the objective of this study and how the collected data will be used. The next section is related to the respondent's profile (e.g., education level and experience). The third section presents the scenario and questions related to it.

* Indicates required question

Survey Consent Statement

1. I hereby give my voluntary and informed consent to participate in the survey titled "Tool support for software maintenance."

*

Purpose: I understand that the purpose of this survey is to collect the perceptions of software practitioners regarding the use of tools for the management of project issues (i.e., issues in issue tracking systems).

Data Usage: I am aware that the data collected through this survey will be used solely for research purposes.

Data Retention: I understand that the data I provide will be stored in an anonymized manner for a non-defined period of time. This means that my individual responses will not be identifiable, and there is no specified time limit for the retention of this data.

Right to Withdraw: I understand that I have the right to stop the survey at any time and request the withdrawal of my answers. My decision to participate in this survey is entirely voluntary, and I may exercise this right without any negative consequences.

By checking the box below, I confirm that I have read and understood the information provided in this consent statement, and I freely consent to participate in the survey.

Check all that apply.

☐ I confirm

Respondent profile

In this section, we invite you to provide some information related to your education and experience in software development.

2. Highest completed academic degree: *

Mark only one oval.

- ☐ Bachelor
- ☐ Master
- ☐ PhD
- ☐ Other: _____

3. Years of experience in software development in industry: *

Mark only one oval.

- ☐ less than 1 year
- ☐ 1 - 5 years
- ☐ 5 - 10 years
- ☐ more than 10 years

4. Current role: *

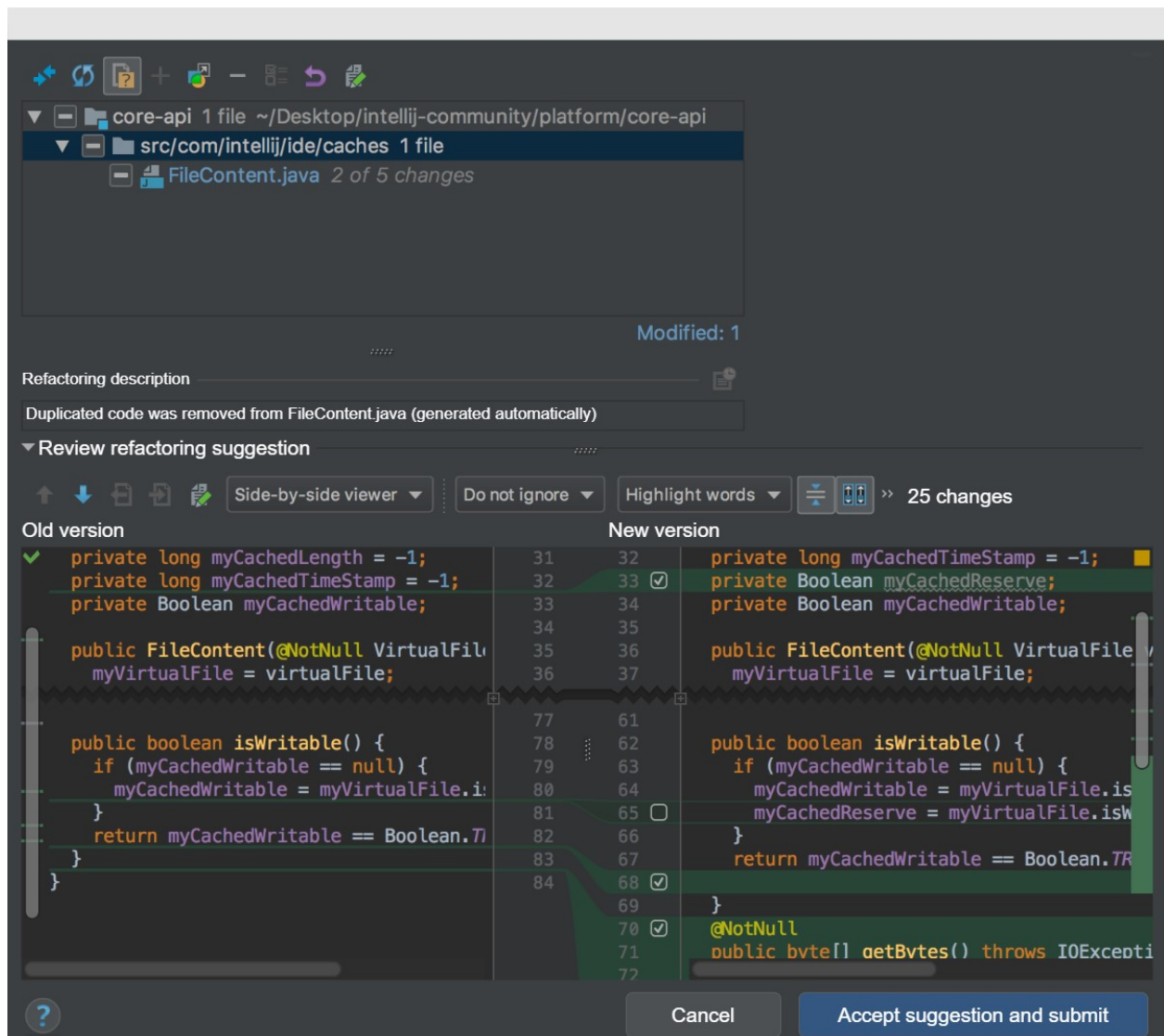
Mark only one oval.

- ☐ Software Architect/Designer
- ☐ Software Developer
- ☐ Project Manager
- ☐ Quality Assurance
- ☐ Other: _____

A IntelliJ IDEA plugin that suggests refactoring actions

In this scenario, a plugin is used to identify code smells in Java code and then suggest refactoring actions, directly inside IntelliJ IDEA. After a refactoring suggestion is presented, the developer can approve the update or not. To increase the level of tool support in this scenario, a Github plugin is used within IDEA to automate the submission of the refactored code to the repository.

See an example of this scenario below.



5. I consider this scenario **useful** for supporting refactoring.

Mark only one oval.

1 2 3 4 5

Strongly ☐ ☐ ☐ ☐ ☐ Strongly Agree

6. Would you prefer a plugin that **proactively** provide refactoring suggestions or do you prefer to execute the plugin **at your own discretion**? Please justify your answer.

7. Would you prefer to use this plugin **within the IDE** or would it be useful if an **external tool** prepared a patch automatically and sent it for review (e.g., to Gerrit)? Please justify your answer.

8. Is there *any other opinion or comment* about this scenario that you would like to provide?

Conclusion

We sincerely appreciate the time you have dedicated to answering our questionnaire. Your responses will greatly support the understanding and evolution of approaches for tool support in software maintenance.

We kindly invite you to participate in a follow-up interview where we can further investigate your perceptions of tool support for software maintenance and analyze other aspects of the scenario.

9. Would you like to participate in a follow-up interview? *

Mark only one oval.

☐ Yes

☐ No

10. If yes, please provide your e-mail for us to contact you.

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